INTERNATIONAL MONETARY FUND

REPUBLIC OF SOUTH SUDAN

FIRST REVIEW UNDER THE STAFF-MONITORED PROGRAM—DEBT SUSTAINABILITY ANALYSIS UPDATE

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Prepared by the staffs of the International Monetary Fund and the International Development Association

Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress	High
Overall risk of debt distress	High
Granularity in the risk rating	Sustainable
Application of judgment	No

South Sudan's debt remains assessed to be sustainable with a high risk of debt distress for both external and overall public debt.[1] Relative to the last assessment in March 2021, the risk of debt distress in this Debt Sustainability Analysis (DSA) has increased despite the new SDR allocation partly used for debt management. This reflects the discovery of higher outstanding oil advances that were contracted in 2018 but not reported to the Ministry of Finance and Planning (MOFP) and thus not included in the March 2021 DSA. There are temporary breaches in three out of seven debt indicators under the baseline scenario (debt service-to-revenues ratio of external public debt, debt service-to-exports ratio of external public debt and present value (PV) of debt-to-GDP ratio of overall public debt), which assumes positive developments in the macro-fiscal framework. These breaches suggest a high risk of external and overall public debt distress. However, all external and overall public debt indicators are expected to be below the respective thresholds from 2026/27 onwards, contingent on robust oil prices and the authorities' commitment to policy adjustment needed to cap the deficit over the medium term together with increased concessional financing. South Sudan's external and overall debt are therefore assessed to be sustainable. Risks to this assessment are tilted to the downside, including in relation to the implementation of policy adjustment, limited access to concessional loans, and oil price volatility. The sustainability of the relative peace and security in the country represent another major risk as the situation in South Sudan continues to be fluid and conflict persists across the country despite recent positive political developments.

[1] South Sudan's debt-carrying capacity remains rated "weak" with composite indicator score of 1.29 according to the April 2021 vintage of the World Economic Outlook and the World Bank's 2019 Country Policy and Institutional Index.

1. Relative to the assessment in March 2021, the outstanding external debt stock and debt service in this DSA update is higher due to an oil advance contracted in 2018 before the first RCF disbursement in November 2020 (Table 1). The newly established Debt Management Unit (DMU) recently discovered an outstanding oil advance of about US$539 million that was contracted in 2018.

2. In the new baseline, South Sudan's debt remains assessed to be sustainable with a high risk of debt distress for both external and overall public debt. In the baseline of the previous DSA, there are temporary breaches in two out of seven debt indicators (debt service-to-revenues ratio of external public debt in FY20/21 to FY23/24 and present value of debt-to-GDP ratio of overall public debt in FY20/21). These breaches lead to a high risk of debt distress. In the baseline of the current DSA, despite of significantly higher oil prices and the use of a portion of new SDR allocation to help close fiscal financing gap, the higher stock of oil advances discussed in ¶1, prolong the breach in the debt-service-to-revenue ratio and the overall public debt to FY25/26.[2] In line with the terms and the fee structure of known oil advances, this DSA update assumes 10-percent effective interest rate on the newly discovered oil advance. Table 2 shows the comparison of key macroeconomic assumptions in this DSA relative to March 2021 DSA, with the main revisions consisting in significantly higher oil prices throughout the forecast period in line with revised WEO projections and a faster convergence of exchange rates in the market relative to what was expected in March 2021.

3. The authorities' strategy is to gradually pay down outstanding oil advances to maintain debt sustainability. The authorities are committed to continue refraining from contracting highly non-concessional debt including new oil advances that have been discontinued following a Council of Ministers Resolution of October 2020, and to gradually reduce the outstanding oil-advance balance. In particular, this DSA assumes that the authorities would reduce the outstanding oil advances by about US$100 million per year in FY2021/22 to FY2022/24 and by about US$200 million per year in FY2024/25 and FY2025/26. The authorities plan to reduce the outstanding amount of oil advance to help create fiscal space given the high effective interest rate[3] and improve fiscal transparency given non-transparent feature of such contract. The authorities expect to fully pay down the outstanding oil advances by June 2026.

4. The authorities are taking steps to strengthen the institutional framework for debt management. As discussed in the accompaying staff report (IMF Country Report No. 2021/xxx), the authorities have tasked the recently reconstituted Loan Committee under the Debt Management Unit of the MOFP to approve any external debt contract by South Sudan before they are sent for final approval by the Council of Ministers and the Assembly. In addition, they have requested TA from the Fund and the World Bank on reducing the cost of servicing debts and formulating a debt management strategy consistent with maintaining public debt sustainability. They have also committed to commissioning an independent audit that will take stock of all outstanding external loan agreements and guarantees by end-2021.

[2] The new DSA baseline utilizes US$150 million of the expected SDR allocation for budget support.

[3] For the 10-percent effective interest rate to hold, the level of oil advance facility should be reduced to the level of outstanding amount on an annual basis. Since some fees are proportional to the level of facility (which can be much larger than the outstanding loan), effective interest rate would be much higher for small amount of outstanding loan but high level of loan facility.

Table 1. Republic of South Sudan: Decomposition of Public Debt by Creditor FY20/21–22/23

	Debt Stock (end of period) Jun-21			Debt Service					
				FY20/21	FY21/22	FY22/23	FY20/21	FY21/22	FY22/23
	(In mn US$)	*(Percent total debt)*	*(Percent GDP)*	*(In mn US$)*			*(Percent GDP)*		
Total	2958	100	60	275	478	453	5.5	7.8	7.2
External[1]	2330	79	47	269	472	447	5.3	7.7	7.1
Multilateral creditors	327	11	7	1	1	2	0.0	0.0	0.0
IMF	227	8	5						
World Bank	82	3	2						
ADB/AfDB/IADB	19	1	0						
Bilateral Creditors	150	5	3	17	19	19	0.3	0.3	0.3
Paris Club	0	0	0	0	1	2	0.0	0.0	0.0
Non-Paris Club	150	5	3	17	18	17	0.3	0.3	0.3
o/w: China EXIM Bank	150	5	3						
Commercial creditors	1853	63	37	251	453	426	5.0	7.4	6.8
o/w: AFREXIM Bank	525	18	11						
QNB	652	22	13						
Oil Companies	676	23	14						
Domestic	627	21	13	6	6	6	0.1	0.1	0.1
Held by residents, total	627	21	13	6	6	6	0.1	0.1	0.1
Held by non-residents, total	0	0	0	0	0	0	0.0	0.0	0.0
Memo items:									
Collateralized debt[2]	1853	63	37						
o/w: Related	0	0	0						
o/w: Unrelated	1853	63	37						
Contingent liabilities	na								
o/w: Public guarantees	na								
o/w: Other explicit contingent liabilities	na								
Nominal GDP (in bn US$)	4969								

1/As reported by Country authorities according to their classification of creditors, including by official and commercial.
2/Debt is collateralized when the creditor has rights over an asset or revenue stream that would allow it, if the borrower defaults on its payment obligations, to rely on the asset or revenue stream to secure repayment of the debt. Collateralization entails a borrower granting liens over specific existing assets or future receivables to a lender as security against repayment of the loan. Collateral is "unrelated" when it has no relationship to a project financed by the loan. An example would be borrowing to finance the budget deficit, collateralized by oil revenue receipts. See the joint IMF-World Bank note for the G20 "Collateralized Transactions: Key Considerations for Public Lenders and Borrowers" for a discussion of issues raised by collateral.

Subsectors of the public sector	Sub-sectors covered
1 Central government	X
2 State and local government	
3 Other elements in the general government	
4 o/w. Social security fund	
5 o/w. Extra budgetary funds (EBFs)	
6 Guarantees (to other entities in the public and private sector, including to SOEs)	
7 Central bank (borrowed on behalf of the government)	X
8 Non-guaranteed SOE debt	

1 The country's coverage of public debt	The central government, central bank		
	Default	Used for the analysis	Reasons for deviations from the default settings
2 Other elements of the general government not captured in 1.	0 percent of GDP	0.0	
3 SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	2.0	
4 PPP	35 percent of PPP stock	0.0	
5 Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5.0	
Total (2+3+4+5) (in percent of GDP)		7.0	

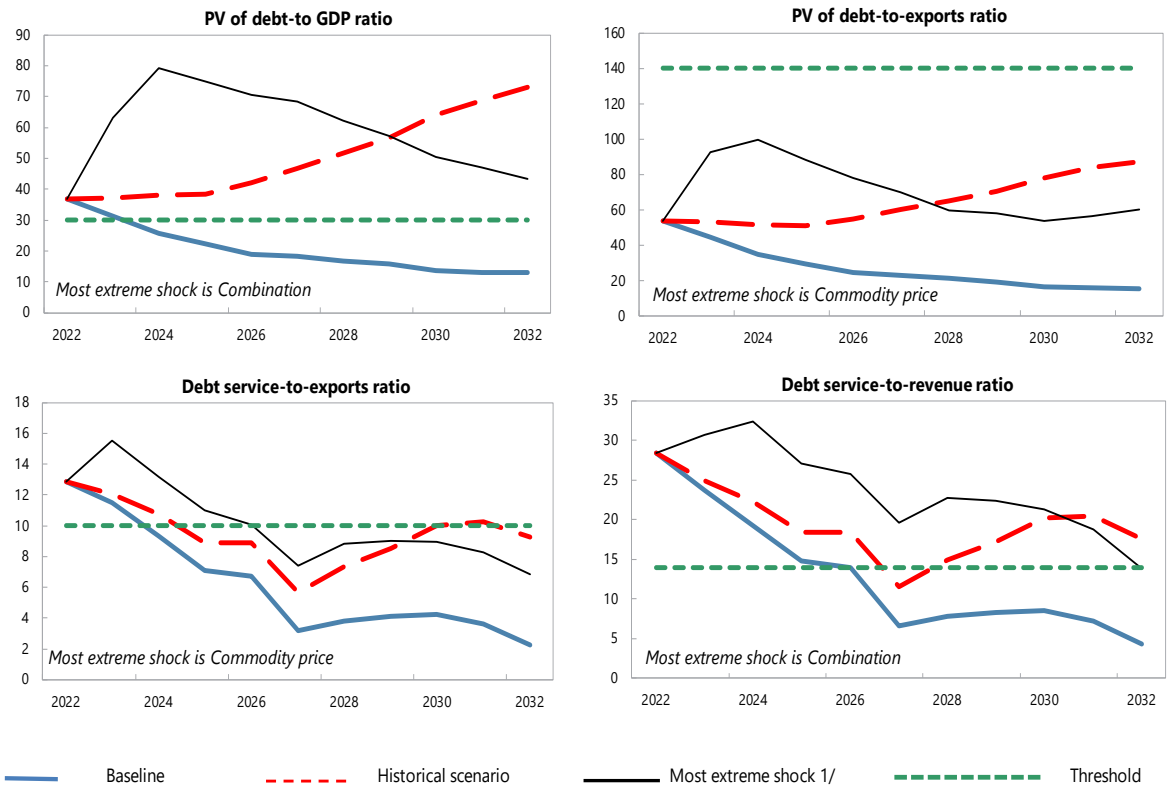
1/ The default shock of 2% of GDP will be triggered for countries, whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

Table 2. Republic of South Sudan: Key Macroeconomic Assumptions Comparison with the Previous DSA

	2019/20	2020/21	2021/22	2022/23	2023/24	2029/30
	Projection					
Real GDP growth (annual percent change)						
Sep 2021 DSA	13.2	-5.4	1.0	5.2	7.4	6.1
Mar 2021 DSA	13.2	-4.2	2.2	2.6	4.7	6.4
Real oil GDP growth (annual percent change)						
Sep 2021 DSA	26.4	-5.9	-2.5	5.8	9.1	5.0
Mar 2021 DSA	26.4	-3.0	3.1	1.2	1.7	5.0
Current Account Balance (percent of GDP)						
Sep 2021 DSA	-2.7	-5.0	2.7	3.2	2.2	5.0
Mar 2021 DSA	-2.7	-6.6	-1.4	0.7	0.6	-1.3
Exports of goods and services (percent of GDP)						
Sep 2021 DSA	64.6	61.8	68.4	70.0	73.5	81.8
Mar 2021 DSA	64.6	61.5	62.9	62.1	61.0	58.7
Imports of goods and services (percent of GDP)						
Sep 2021 DSA	79.5	81.8	75.8	83.6	89.8	100.6
Mar 2021 DSA	79.5	84.4	76.8	81.0	81.8	82.2
Primary deficit (percent of GDP)						
Sep 2021 DSA	-5.1	-4.6	4.7	3.4	2.9	0.8
Mar 2021 DSA	-5.1	-1.9	-1.0	1.2	1.4	0.6
Revenue and grants (percent of GDP)						
Sep 2021 DSA	29.7	31.6	29.6	31.9	33.0	33.2
Mar 2021 DSA	29.7	24.6	31.2	31.3	30.9	31.7
Primary expenditures (percent of GDP)						
Sep 2021 DSA	34.8	36.2	24.9	28.5	30.1	32.5
Mar 2021 DSA	34.8	26.5	32.2	30.1	29.5	31.2
SSD Oil prices (US$/ barrel)						
Sep 2021 DSA	49.3	51.5	63.1	63.2	64.5	72.9
Mar 2021 DSA	49.3	47.9	54.7	53.4	54.5	61.6

Sources: South Sudanese authorities; and IMF staff estimations and projections.

Figure 1. Republic of South Sudan: Indicators of Public and Publicly Guaranteed External Debt Under Alternatives Scenarios, FY2022–32[1/]



Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	No	
Natural Disasters	n.a.	n.a.
Commodity Prices [2/]	Yes	No
Market Financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	100%
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	4.1%	4.1%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	16	16
Avg. grace period	3	3

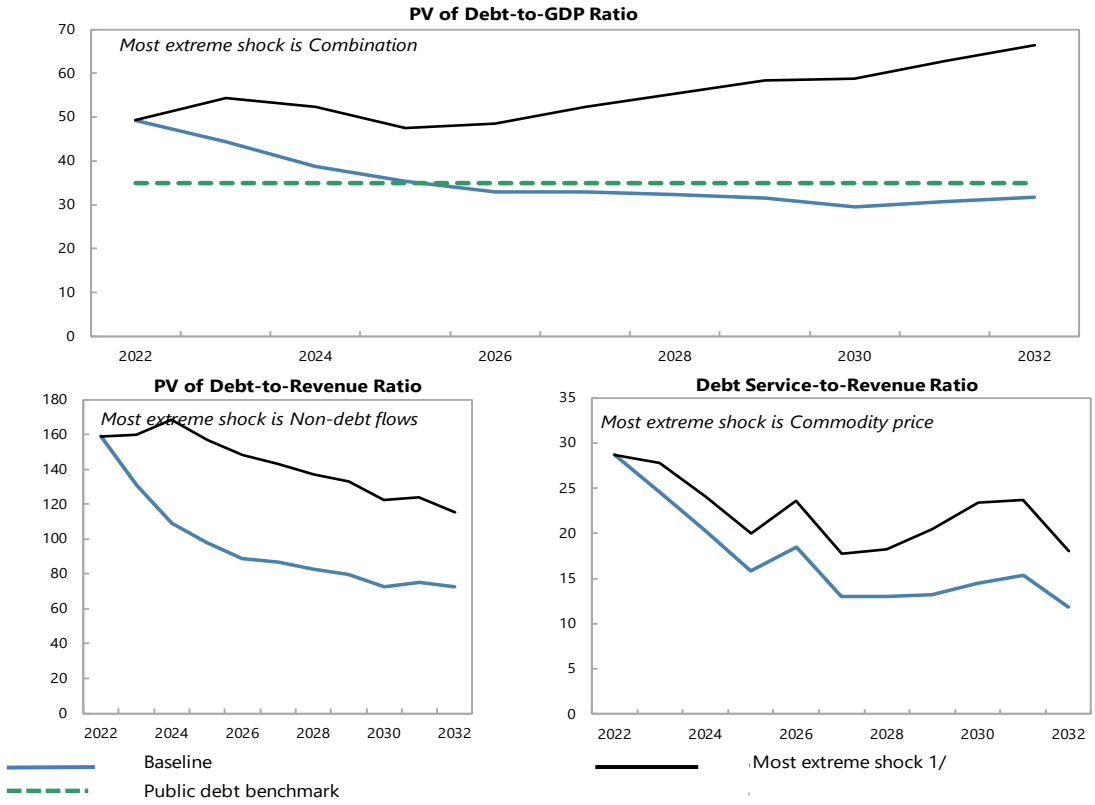
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2032. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Republic of South Sudan: Indicators of Public Debt Under Alternative Scenarios, FY2022–32



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	38%	38%
Domestic medium and long-term	62%	62%
Domestic short-term	0%	0%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	4.1%	4.1%
Avg. maturity (incl. grace period)	16	16
Avg. grace period	3	3
Domestic MLT debt		
Avg. real interest rate on new borrowing	-9.0%	2.0%
Avg. maturity (incl. grace period)	5	5
Avg. grace period	3	3
Domestic short-term debt		
Avg. real interest rate	0%	0%

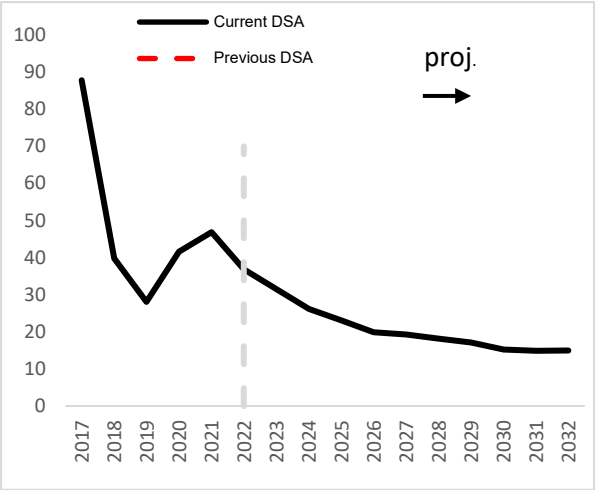
* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in or before 2032. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

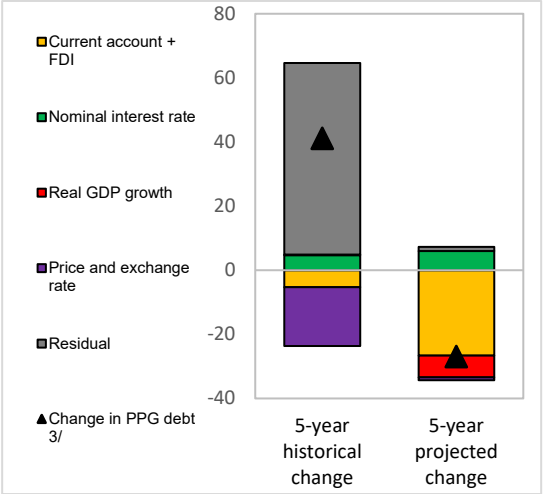
Figure 3. Republic of South Sudan: Drivers of Debt Dynamics—Baseline Scenario[1/]

External Debt

Gross Nominal PPG External Debt 2/
(in percent of GDP; DSA vintages)

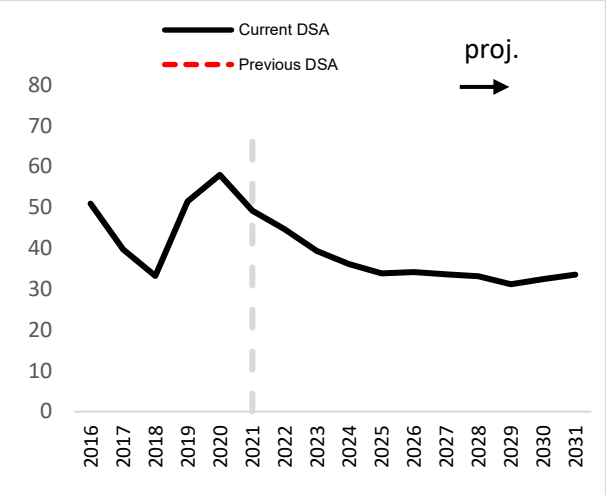


Debt-Creating Flows
(percent of GDP)

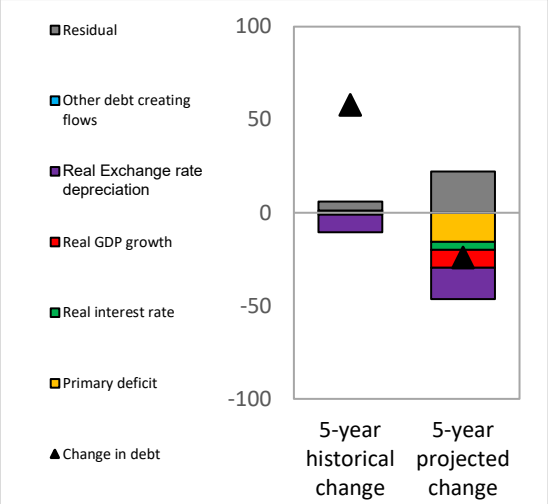


Public Debt

Gross Nominal Public Debt 2/
(in percent of GDP; DSA vintages)



Debt-Creating flows
(percent of GDP)

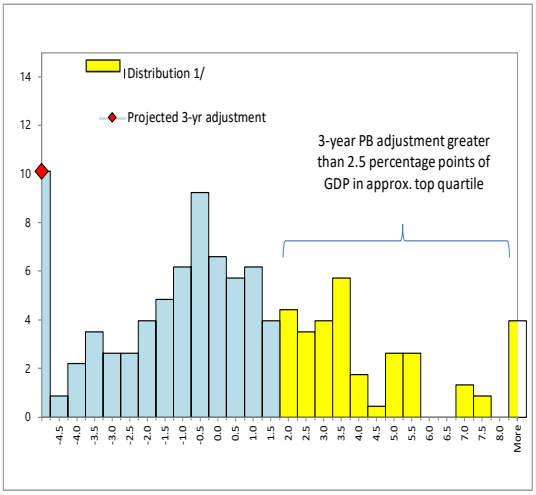


1/ Analyses on unexpected changes in debt are unavailable due to the lack of data.

2/ The current DSA assumes more external financing for peace process than the previous DSA, which makes its Gross Nominal PPG External Debt and Gross Nominal Public Debt larger than the previous DSA.

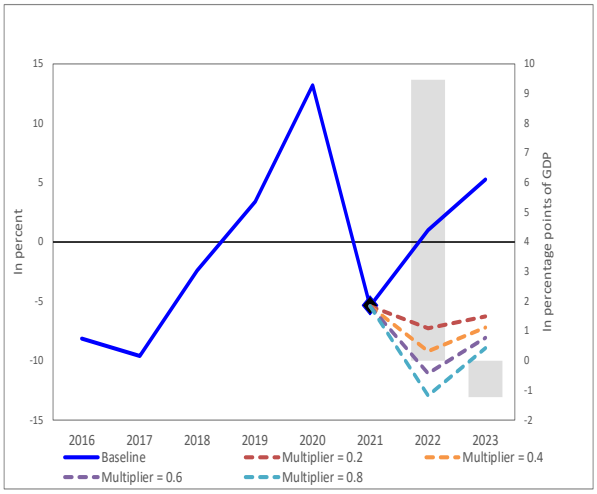
Figure 4. Republic of South Sudan: Realism Tools

3-Year Adjustment in Primary Balance (Percentage points of GDP)



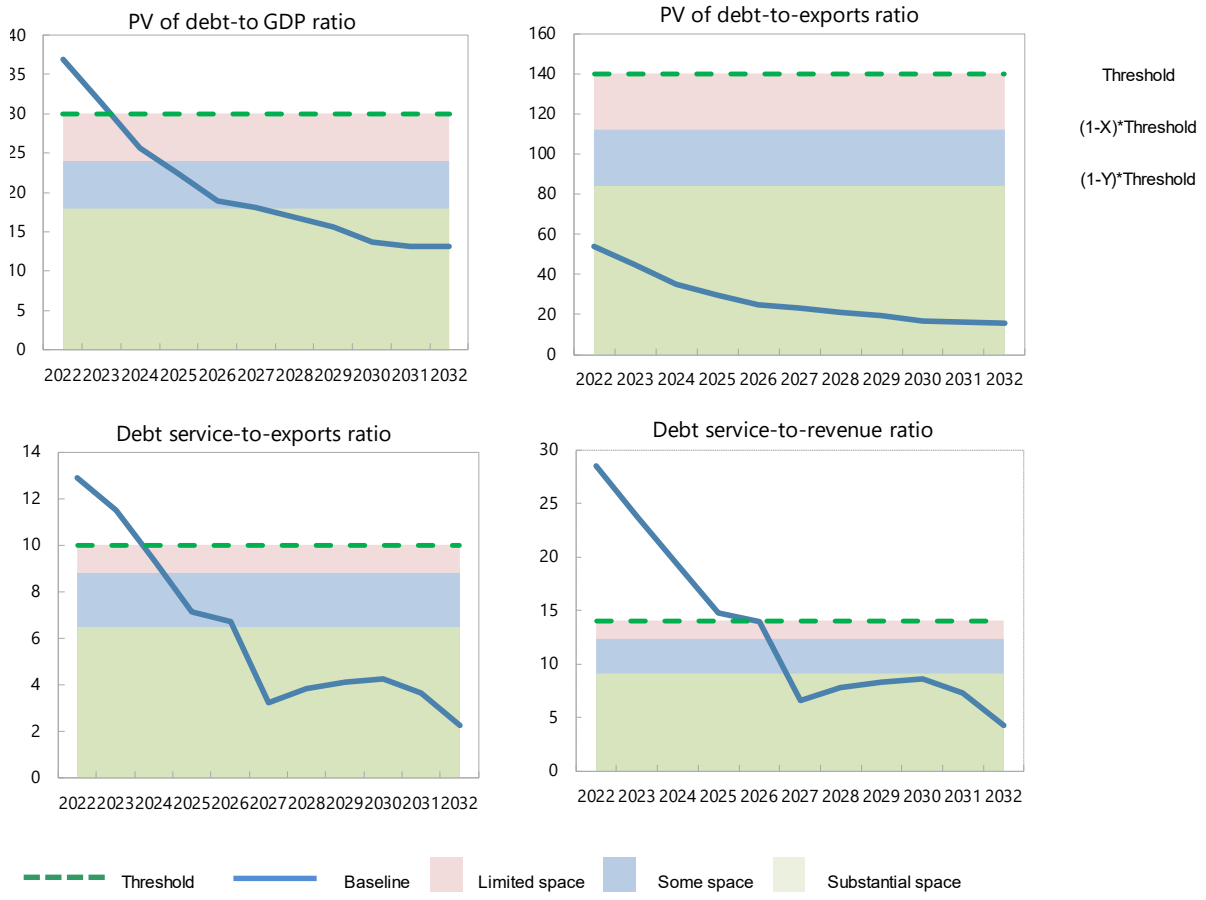
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Figure 5. Republic of South Sudan: Qualification of the Moderate Category. FY2022–32[1/]



Sources: Country authorities; and staff estimates and projections.

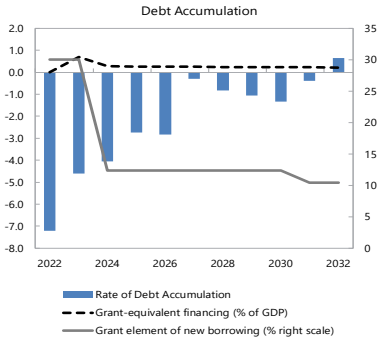
1/ For the PV debt/GDP and PV debt/exports thresholds, x is 20 percent and y is 40 percent. For debt service/Exports and debt service/revenue thresholds, x is 12 percent and y is 35 percent.

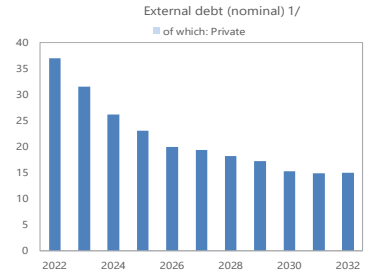
Table 3. Republic of South Sudan: External Debt Sustainability Framework, Baseline Scenario, FY2021–42

(In percent of GDP, unless otherwise indicated)

	Actual	Projections								Average 8/	
Fiscal Year 9/	2021	2022	2023	2024	2025	2026	2027	2032	2042	Historical	Projections
External debt (nominal) 1/	**46.9**	**36.9**	**31.5**	**26.2**	**23.1**	**19.9**	**19.3**	**15.0**	**13.8**	**25.4**	**21.6**
of which: public and publicly guaranteed (PPG)	**46.9**	**36.9**	**31.5**	**26.2**	**23.1**	**19.9**	**19.3**	**15.0**	**13.8**	**25.4**	**21.6**
Change in external debt	5.3	-10.0	-5.4	-5.3	-3.1	-3.1	-0.6	0.1	-7.4		
Identified net debt-creating flows	**3.6**	**-4.0**	**-5.9**	**-6.3**	**-3.9**	**-7.3**	**-5.7**	**-3.9**	**-2.8**	**-4.3**	**-5.8**
Non-interest current account deficit	**2.9**	**-4.8**	**-4.8**	**-3.3**	**-0.4**	**-4.3**	**-2.4**	**5.0**	**5.5**	**2.6**	**-2.8**
Deficit in balance of goods and services	20.1	7.3	13.7	16.3	20.3	16.6	19.4	36.2	36.0	**10.3**	**18.4**
Exports	61.8	68.4	70.0	73.5	74.9	76.4	77.8	83.3	46.5		
Imports	81.8	75.8	83.6	89.8	95.2	93.0	97.2	119.5	82.6		
Net current transfers (negative = inflow)	-25.9	-23.0	-29.4	-29.7	-30.5	-31.2	-31.9	-37.7	-36.5	**-22.4**	**-31.3**
of which: official	0.0	-0.3	-0.5	-0.5	-0.5	-0.5	-0.5	-0.4	0.0		
Other current account flows (negative = net inflow)	8.7	10.9	11.0	10.2	9.7	10.3	10.0	6.5	5.9	**14.6**	**10.1**
Net FDI (negative = inflow)	**-0.9**	**-0.9**	**-0.8**	**-1.9**	**-3.0**	**-2.5**	**-2.8**	**-8.7**	**-3.3**	**0.7**	**-2.8**
Endogenous debt dynamics 2/	**1.6**	**1.7**	**-0.3**	**-1.1**	**-0.5**	**-0.5**	**-0.5**	**-0.2**	**-5.0**		
Contribution from nominal interest rate	2.1	2.1	1.5	1.1	0.8	0.5	0.4	0.5	0.6		
Contribution from real GDP growth	2.2	-0.4	-1.9	-2.2	-1.2	-1.0	-0.9	-0.7	-5.6		
Contribution from price and exchange rate changes	-2.7	...	...	...	...	...	...	...	...		
Residual 3/	**1.8**	**-6.0**	**0.5**	**1.0**	**0.8**	**4.2**	**5.1**	**4.0**	**-4.6**	**11.4**	**2.6**
of which: exceptional financing	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators											
PV of PPG external debt-to-GDP ratio	**46.9**	**36.9**	**31.2**	**25.6**	**22.3**	**18.9**	**18.1**	**13.1**	**12.4**		
PV of PPG external debt-to-exports ratio	**75.9**	**53.9**	**44.6**	**34.9**	**29.7**	**24.8**	**23.3**	**15.8**	**26.6**		
PPG debt service-to-exports ratio	**8.3**	**12.9**	**11.5**	**9.3**	**7.1**	**6.7**	**3.2**	**2.3**	**2.5**		
PPG debt service-to-revenue ratio	**16.2**	**28.4**	**23.7**	**19.3**	**14.7**	**13.9**	**6.6**	**4.3**	**4.3**		
Gross external financing need (Million of U.S. dollars)	352.6	167.9	137.9	97.3	115.2	-101.3	-175.2	-131.3	463.9		
Key macroeconomic assumptions											
Real GDP growth (in percent)	-5.4	1.0	5.2	7.4	4.9	4.6	4.5	5.3	36.9	**5.2**	**4.6**
GDP deflator in US dollar terms (change in percent)	7.1	6.5	-1.7	-1.4	-2.0	-1.8	-1.7	3.0	3.0	**-4.6**	**-0.3**
Effective interest rate (percent) 4/	5.2	4.8	4.3	3.6	3.0	2.3	2.0	3.7	3.9	**2.2**	**3.4**
Growth of exports of G&S (US dollar terms, in percent)	-1.7	19.2	5.8	11.2	4.9	4.8	4.7	6.9	-14.1	**204.8**	**8.4**
Growth of imports of G&S (US dollar terms, in percent)	-7.0	-0.4	14.3	13.7	9.1	0.4	7.5	13.0	-2.2	**6.2**	**7.4**
Grant element of new public sector borrowing (in percent)	...	30.0	30.0	12.4	12.4	12.4	12.4	10.4	7.9	**...**	**18.3**
Government revenues (excluding grants, in percent of GDP)	31.6	31.0	34.0	35.6	36.2	37.0	38.1	43.7	27.4	**33.8**	**35.3**
Aid flows (in Million of US dollars) 5/	0.0	0.0	30.0	30.0	30.0	30.0	30.0	30.0	30.0		
Grant-equivalent financing (in percent of GDP) 6/	...	0.0	0.7	0.3	0.3	0.3	0.3	0.2	0.1	**...**	**0.3**
Grant-equivalent financing (in percent of external financing) 6/	...	...	30.0	12.4	12.4	12.4	12.4	10.4	7.9	**...**	**15.9**
Nominal GDP (Million of US dollars)	4,969	5,344	5,531	5,858	6,024	6,192	6,363	7,322	13,866		
Nominal dollar GDP growth	1.3	7.5	3.5	5.9	2.8	2.8	2.8	4.7	63.5	**-2.6**	**4.2**
Memorandum items:											
PV of external debt 7/	46.9	36.9	31.2	25.6	22.3	18.9	18.1	13.1	12.4		
In percent of exports	75.9	53.9	44.6	34.9	29.7	24.8	23.3	15.8	26.6		
Total external debt service-to-exports ratio	8.3	12.9	11.5	9.3	7.1	6.7	3.2	2.3	2.5		
PV of PPG external debt (in Million of US dollars)	2330.4	1972.4	1726.2	1502.0	1342.0	1171.3	1153.3	961.7	1714.2		
(PVt-PVt-1)/GDPt-1 (in percent)		-7.2	-4.6	-4.1	-2.7	-2.8	-0.3	0.6	1.2		
Non-interest current account deficit that stabilizes debt ratio	-2.4	5.2	0.6	2.0	2.7	-1.1	-1.8	4.9	12.9		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

9/ The fiscal year runs from July to June, i.e., 2021 refers to FY20/21 ending in June 2021.

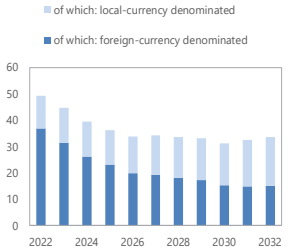
Table 4. Republic of South Sudan: Public Sector Debt Sustainability Framework, Baseline Scenario, FY2021–42

(In percent of GDP, unless otherwise indicated)

	Actual	Projections								Average 6/	
Fiscal Year 7/	2021	2022	2023	2024	2025	2026	2027	2032	2042	Historical	Projections
Public sector debt 1/	**58.0**	**49.2**	**44.6**	**39.4**	**36.2**	**33.9**	**34.2**	**33.5**	**32.4**	**25.3**	**36.5**
of which: external debt	**46.9**	**36.9**	**31.5**	**26.2**	**23.1**	**19.9**	**19.3**	**15.0**	**13.8**	**20.7**	**21.6**
Change in public sector debt	6.5	-8.8	-4.6	-5.3	-3.2	-2.3	0.3	1.1	-18.4		
Identified debt-creating flows	**17.7**	**-30.9**	**-4.6**	**-5.3**	**-3.2**	**-2.3**	**0.3**	**1.1**	**-18.7**	**4.5**	**-4.2**
Primary deficit	**-10.2**	**-4.9**	**-3.7**	**-3.1**	**-2.2**	**-1.7**	**0.9**	**1.9**	**0.3**	**-4.3**	**-1.2**
Revenue and grants	31.6	31.0	34.0	35.6	36.2	37.0	38.1	43.7	27.4	**38.8**	**37.8**
of which: grants	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Primary (noninterest) expenditure	21.4	26.1	30.3	32.5	34.1	35.3	39.0	45.6	27.7	**34.5**	**36.6**
Automatic debt dynamics	**28.0**	**-26.0**	**-0.9**	**-2.2**	**-1.1**	**-0.6**	**-0.6**	**-0.8**	**-19.0**		
Contribution from interest rate/growth differential	2.6	-5.1	-2.2	-3.2	-2.1	-1.4	-1.3	-1.2	-16.3		
of which: contribution from average real interest rate	-0.3	-4.5	0.2	-0.1	-0.2	0.2	0.1	0.4	-2.6		
of which: contribution from real GDP growth	2.9	-0.6	-2.4	-3.1	-1.8	-1.6	-1.5	-1.6	-13.7		
Contribution from real exchange rate depreciation	25.3	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**-11.2**	**1.2**	**1.3**	**1.0**	**1.0**	**0.9**	**0.7**	**0.4**	**-2.3**	**1.6**	**0.8**
Sustainability indicators											
PV of public debt-to-GDP ratio 2/	**58.0**	**49.2**	**44.4**	**38.8**	**35.4**	**32.9**	**33.0**	**31.6**	**31.0**		
PV of public debt-to-revenue and grants ratio	**183.3**	**158.9**	**130.5**	**109.1**	**97.7**	**88.8**	**86.5**	**72.4**	**112.9**		
Debt service-to-revenue and grants ratio 3/	**16.5**	**28.7**	**24.5**	**20.3**	**15.9**	**18.5**	**13.0**	**11.9**	**19.3**		
Gross financing need 4/	5.8	4.0	4.7	4.1	3.6	5.2	5.8	7.1	5.6		
Key macroeconomic and fiscal assumptions											
Real GDP growth (in percent)	-5.4	1.0	5.2	7.4	4.9	4.6	4.5	5.3	36.9	**5.2**	**4.5**
Average nominal interest rate on external debt (in percent)	5.3	4.8	4.3	3.6	3.0	2.3	2.0	3.7	3.9	**3.7**	**3.2**
Average real interest rate on domestic debt (in percent)	4.6	3.3	2.4	1.7	1.1	0.4	0.0	1.6	3.9	**1.0**	**1.3**
Real exchange rate depreciation (in percent, + indicates depreciation)	56.2	...	...	...	...	...	...	...	...	**66.8**	
Inflation rate (GDP deflator, in percent)	27.0	123.4	8.2	8.5	7.8	3.1	3.2	4.4	25.4	**82.6**	**15.6**
Growth of real primary spending (deflated by GDP deflator, in percent)	-8.3	-27.3	22.4	15.1	10.0	8.4	15.5	14.7	-24.7	**109.8**	**7.5**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-16.8	3.9	0.9	2.2	1.1	0.6	0.6	0.8	18.7	**-6.9**	**1.0**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Currency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/



Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government, central bank. Definition of external debt is Currency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

7/ The fiscal year runs from July to June, i.e., 2021 refers to FY20/21 ending in June 2021.

Table 5. Republic of South Sudan: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, FY2022–32

(in percent)

	Projections 1/										
	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
PV of debt-to GDP ratio											
Baseline	**36.9**	**31.2**	**25.6**	**22.3**	**18.9**	**18.1**	**16.8**	**15.6**	**13.6**	**13.1**	**13.1**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2042 2/	**36.9**	**37.2**	**37.9**	**38.4**	**42.1**	**46.8**	**51.8**	**56.6**	**64.1**	**68.6**	**72.9**
B. Bound Tests											
B1. Real GDP growth	**36.9**	**39.5**	**38.1**	**33.1**	28.1	26.9	25.0	23.4	20.7	20.0	19.8
B2. Primary balance	**36.9**	**34.8**	**32.8**	29.9	26.8	26.8	26.8	26.5	24.2	23.9	24.1
B3. Exports	**36.9**	**38.2**	**44.4**	**41.3**	**38.3**	**37.4**	**34.2**	**31.6**	27.9	26.1	24.4
B4. Other flows 3/	**36.9**	**41.1**	**46.8**	**43.8**	**40.8**	**39.7**	**36.3**	**33.5**	29.5	27.6	25.7
B5. One-time 30 percent nominal depreciation	**36.9**	**39.7**	26.7	22.3	17.9	16.8	15.8	14.8	13.1	13.0	13.3
B6. Combination of B1-B5	**36.9**	**63.2**	**79.3**	**74.9**	**70.6**	**68.2**	**62.3**	**57.3**	**50.5**	**46.9**	**43.3**
C. Tailored Tests											
C1. Combined contingent liabilities	**36.9**	**33.7**	28.2	24.9	21.6	21.4	20.6	19.6	17.6	17.4	17.6
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	**36.9**	**48.7**	**58.9**	**56.1**	**52.9**	**50.6**	**45.8**	**42.5**	**38.6**	**38.8**	**40.7**
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**
PV of debt-to-exports ratio											
Baseline	53.9	44.6	34.9	29.7	24.8	23.3	21.2	19.5	16.7	16.1	15.8
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2042 2/	53.9	53.1	51.6	51.3	55.1	60.1	65.3	70.7	78.3	84.0	87.5
B. Bound Tests											
B1. Real GDP growth	53.9	44.6	34.9	29.7	24.8	23.3	22.3	21.6	19.5	19.8	19.9
B2. Primary balance	53.9	49.8	44.7	39.9	35.0	34.4	33.8	33.0	29.6	29.3	28.9
B3. Exports	53.9	58.9	72.2	65.9	59.8	57.3	54.1	51.9	46.7	45.9	43.5
B4. Other flows 3/	53.9	58.8	63.6	58.4	53.4	51.0	48.1	46.0	41.4	40.6	38.3
B5. One-time 30 percent nominal depreciation	53.9	44.6	28.6	23.4	18.4	17.0	16.4	16.0	14.4	15.0	15.6
B6. Combination of B1-B5	53.9	77.2	71.1	80.0	73.9	70.1	65.9	63.0	56.6	55.2	51.7
C. Tailored Tests											
C1. Combined contingent liabilities	53.9	48.1	38.4	33.3	28.3	27.5	26.0	24.5	21.5	21.3	21.1
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	53.9	92.6	99.9	88.5	78.1	70.3	60.0	57.9	53.6	56.6	60.1
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**
Debt service-to-exports ratio											
Baseline	**12.9**	**11.5**	9.3	7.1	6.7	3.2	3.8	4.1	4.3	3.6	2.3
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2042 2/	**12.9**	**12.1**	**10.8**	8.9	8.9	5.6	7.4	8.5	**10.0**	**10.2**	9.2
B. Bound Tests											
B1. Real GDP growth	**12.9**	**11.5**	9.3	7.1	6.7	3.2	3.8	4.2	4.4	3.7	2.5
B2. Primary balance	**12.9**	**11.5**	9.5	7.5	7.2	4.0	5.0	5.4	5.5	5.0	3.7
B3. Exports	**12.9**	**12.7**	**11.7**	9.9	9.4	5.8	7.9	8.3	8.3	7.6	5.9
B4. Other flows 3/	**12.9**	**11.5**	9.9	8.3	7.9	5.3	6.9	7.3	7.3	6.7	5.3
B5. One-time 30 percent nominal depreciation	**12.9**	**11.5**	9.3	6.9	6.5	2.9	3.1	3.5	3.7	3.1	1.9
B6. Combination of B1-B5	**12.9**	**12.7**	**12.5**	**10.5**	10.0	7.7	9.4	9.7	9.7	9.0	7.2
C. Tailored Tests											
C1. Combined contingent liabilities	**12.9**	**11.5**	9.5	7.3	6.9	3.4	4.0	4.3	4.5	3.8	2.5
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	**12.9**	**15.5**	**13.2**	**11.0**	**10.1**	7.4	8.8	9.0	8.9	8.3	6.8
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**10**	**10**	**10**	**10**	**10**	**10**	**10**	**10**	**10**	**10**	**10**
Debt service-to-revenue ratio											
Baseline	**28.4**	**23.7**	**19.3**	**14.7**	13.9	6.6	7.8	8.3	8.6	7.2	4.3
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2042 2/	**28.4**	**24.9**	**22.2**	**18.4**	**18.4**	11.5	**15.0**	**17.2**	**20.1**	**20.4**	**17.6**
B. Bound Tests											
B1. Real GDP growth	**28.4**	**30.0**	**28.7**	**21.9**	**20.7**	9.7	11.0	11.4	11.3	9.3	5.6
B2. Primary balance	**28.4**	**23.7**	**19.7**	**15.6**	**14.8**	8.1	10.1	10.8	11.1	9.9	7.1
B3. Exports	**28.4**	**24.2**	**20.3**	**17.1**	**16.2**	9.9	12.8	12.7	12.2	10.6	7.6
B4. Other flows 3/	**28.4**	**23.7**	**20.4**	**17.2**	**16.3**	10.8	13.4	13.3	12.8	11.1	8.1
B5. One-time 30 percent nominal depreciation	**28.4**	**30.1**	**24.5**	**18.0**	**17.0**	7.7	7.7	8.1	8.2	6.5	3.6
B6. Combination of B1-B5	**28.4**	**30.6**	**32.4**	**27.0**	**25.7**	**19.6**	**22.7**	**22.4**	**21.3**	**18.7**	13.9
C. Tailored Tests											
C1. Combined contingent liabilities	**28.4**	**23.7**	**19.6**	**15.0**	**14.2**	6.8	8.1	8.7	9.0	7.6	4.7
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	**28.4**	**26.2**	**23.7**	**20.9**	**19.6**	**14.5**	**17.6**	**16.7**	**15.8**	13.9	10.6
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**14**	**14**	**14**	**14**	**14**	**14**	**14**	**14**	**14**	**14**	**14**

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 6. Republic of South Sudan: Sensitivity Analysis for Key Indicators of Public Debt FY2022–32

	Projections 1/										
	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
PV of Debt-to-GDP Ratio											
Baseline	**49**	**44.4**	**38.8**	**35.4**	32.9	33.0	32.2	31.6	29.5	30.6	31.6
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2042 2/	**49**	25	13	6	3	2	1	1	0	0	0
B. Bound Tests											
B1. Real GDP growth	**49**	**52**	**53**	**48**	**45**	**45**	**44**	**43**	**40**	**42**	**43**
B2. Primary balance	**49**	**54**	**58**	**55**	**52**	**52**	**51**	**51**	**49**	**50**	**51**
B3. Exports	**49**	**50**	**57**	**54**	**52**	**52**	**51**	**50**	**47**	**48**	**48**
B4. Other flows 3/	**49**	**54**	**60**	**57**	**55**	**55**	**54**	**53**	**50**	**51**	**50**
B5. One-time 30 percent nominal depreciation	**49**	**49**	**41**	35	29	26	23	19	14	11	9
B6. Combination of B1-B5	**49**	**54**	**52**	**48**	**48**	**52**	**55**	**58**	**59**	**63**	**66**
C. Tailored Tests											
C1. Combined contingent liabilities	**49**	**51**	**45**	**42**	**39**	**39**	**39**	**38**	**36**	**37**	**38**
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	**49**	**49**	**48**	**48**	**49**	**52**	**52**	**51**	**49**	**50**	**51**
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Public debt benchmark	**35**	**35**	**35**	**35**	**35**	**35**	**35**	**35**	**35**	**35**	**35**
PV of Debt-to-Revenue Ratio											
Baseline	158.9	130.5	109.1	97.7	88.8	86.5	82.7	79.7	72.6	74.9	72.4
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2042 2/	159	73	35	18	9	4	2	1	1	1	1
B. Bound Tests											
B1. Real GDP growth	159	130	109	98	89	87	83	80	73	75	72
B2. Primary balance	159	160	163	150	141	137	132	129	119	122	116
B3. Exports	159	149	160	148	139	136	130	127	116	118	111
B4. Other flows 3/	159	160	168	157	148	143	137	133	122	124	115
B5. One-time 30 percent nominal depreciation	159	144	114	95	79	69	58	47	34	27	20
B6. Combination of B1-B5	159	160	147	131	131	137	142	147	144	153	152
C. Tailored Tests											
C1. Combined contingent liabilities	159	151	128	116	106	103	99	96	88	91	87
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	159	157	145	145	141	141	135	129	120	123	117
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	28.7	24.5	20.3	15.9	18.5	13.0	13.0	13.2	14.5	15.3	11.9
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2042 2/	29	13	6	2	1	1	0	0	0	0	0
B. Bound Tests											
B1. Real GDP growth	29	25	20	16	18	13	13	13	14	15	12
B2. Primary balance	29	25	23	20	21	22	28	23	20	22	20
B3. Exports	29	25	21	18	21	16	18	19	20	21	17
B4. Other flows 3/	29	25	21	18	21	17	19	19	21	21	18
B5. One-time 30 percent nominal depreciation	29	25	23	17	20	12	12	11	13	12	8
B6. Combination of B1-B5	29	26	24	19	22	16	17	21	26	28	24
C. Tailored Tests											
C1. Combined contingent liabilities	29	25	22	17	19	17	17	14	16	17	14
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	29	28	24	20	24	18	18	20	23	24	18
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.

1/ A bold value indicates a breach of the benchmark.

2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.

3/ Includes official and private transfers and FDI.